

Requirement Specification Document

Project Overview

This document outlines the requirements for the functional, UI, performance, security, accessibility, and SEO testing of the website. These requirements serve as the foundation for creating test cases, ensuring comprehensive coverage and validation of the system's behavior and quality.

1. Functional Requirements

- The website must provide fully functional internal and external links, including those in the menu bar, page content, and footer. All links should navigate correctly to the intended pages or external sites without errors.
 - External links should open in a new browser tab or window to improve user experience and prevent navigation away from the main site.
 - The system shall display a custom 404 error page for any invalid or incorrect URLs entered by the user, ensuring a user-friendly error handling process.
 - The contact form on the website must be clearly visible and functional, including all required fields. The form should validate inputs correctly, such as email addresses accepting only valid formats and mandatory fields not allowing empty submissions. The submit button must correctly submit the form data.
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2. User Interface (UI) and User Experience (UX) Requirements

- The website layout and design should be consistent across all pages, maintaining uniform font colors, background colors, and overall visual elements to provide a seamless user experience.
 - The website must be responsive, adapting flawlessly to different device types and screen sizes, including mobile phones, tablets, and desktops. This responsiveness must extend to all elements including navigation menus and forms.
 - Alt tags must be provided for all images to support accessibility, and color contrast of text and background should meet readability standards for users with visual impairments.
 - Favicons located in the footer area should be functional and clickable, enhancing the site's branding and user navigation.
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3. Performance Requirements

- The total page load time must be optimized to load within an acceptable threshold to ensure fast and efficient access to site content.
 - Key performance metrics such as DOM content load time, image load time, and resource load time (for CSS, JavaScript, etc.), Largest Contentful Paint (LCP), Cumulative Layout Shift (CLS), and Interaction to Next Paint (INP) must meet predefined targets to guarantee a smooth user experience without delays or layout shifts.
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4. Security Requirements

- The website must utilize HTTPS protocol with a valid security certificate to protect user data and secure all communications.
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5. Compatibility Requirements

- The website should be fully compatible with all major web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, ensuring consistent behavior and appearance across platforms.
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6. SEO and Metadata Requirements

- Proper meta tags must be included on all pages to enhance search engine optimization (SEO). These include meta descriptions, indexing instructions (index/follow), and social media sharing tags.
 - The content on each page must be relevant to the page title and should not contain spelling or grammatical errors to maintain professionalism and improve SEO effectiveness.
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7. Technical Requirements

- Pages should use the correct character encoding (UTF-8) to avoid text display issues across browsers and platforms.
- Meta tags must also support responsiveness and device adaptability, further aiding in mobile usability and SEO compliance.